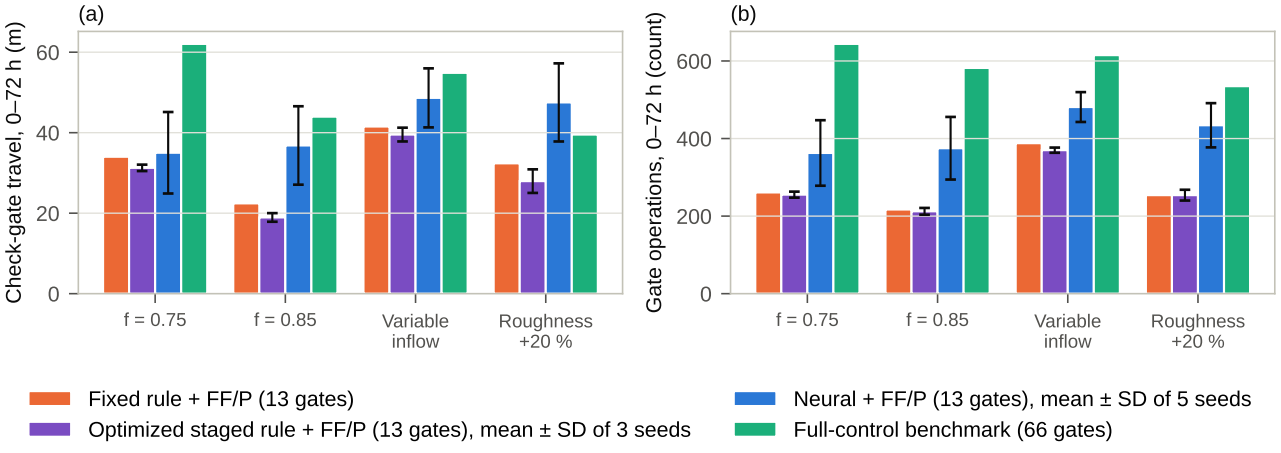
(a)
Check-gate travel, 0–72 h (m)
0
20
40
60
f = 0.75
f = 0.85
Variable inflow
Roughness +20 %
(b)
Gate operations, 0–72 h (count)
0
200
400
600
f = 0.75
f = 0.85
Variable inflow
Roughness +20 %
Fixed rule + FF/P (13 gates)
Optimized staged rule + FF/P (13 gates), mean ± SD of 3 seeds
Neural + FF/P (13 gates), mean ± SD of 5 seeds
Full-control benchmark (66 gates)